Predicting Heart Disease Risk

A Multi-Model Classification Approach

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1. Introduction

Heart disease remains the leading cause of death globally, yet early detection can significantly improve patient outcomes. This study addresses a fundamental clinical question: *Which patient characteristics most effectively predict heart disease, and how can we build interpretable models for clinical screening?*

We employ a multi-model approach, comparing logistic regression and decision trees for classification while using PCA for dimensionality reduction and K-means clustering for patient segmentation. This combination allows us to maximize predictive accuracy while maintaining clinical interpretability.

1.1 Dataset Overview

Our dataset comprises 918 patient records with 11 predictor variables spanning three categories:

- **Demographics** (Age, Sex)
- **Clinical measurements** (RestingBP, Cholesterol, FastingBS, MaxHR, Oldpeak)
- **Diagnostic indicators** (ChestPainType, RestingECG, ExerciseAngina, ST_Slope).

The binary target variable indicates presence or absence of heart disease. We used stratified sampling to create a 70/30 train-test split (n=643 training, n=275 testing), preserving class proportions across both sets.

2. Methodology

2.1 Data Preprocessing

Categorical variables were encoded using full rank encoding to prevent multicollinearity, expanding our feature space to 15 predictors. Continuous features were z-score standardized, with transformation parameters learned exclusively from training data to prevent information leakage. This standardization is essential for PCA and ensures equal feature weighting in clustering.

2.2 Principal Component Analysis

PCA transformed our 15 correlated features into orthogonal components. Applying the 80% cumulative variance threshold, we retained **k=9 principal components**. Analysis of loadings revealed interpretable clinical factors: PC1 ('Ischemia and Physical Stress') was driven by ST_SlopeFlat and ExerciseAnginaY, while PC2 ('ECG Health Status') distinguished normal from abnormal resting ECG readings. Together, these two components captured over 33% of total variance.

2.3 Classification Models

Logistic Regression: We developed two models: a Benchmark using all 15 raw features, and a PCA-Enhanced model using 9 principal components. VIF analysis revealed ST_Slope variables approaching VIF=5 in the Benchmark model, validating our decision to use PCA for coefficient stability.

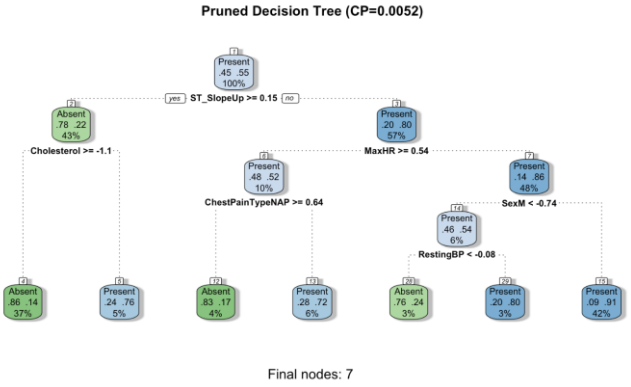


Figure 1. Pruned Decision Tree

Decision Tree: A classification tree was grown without constraints, then pruned using 10-fold cross-validation to select the optimal complexity parameter (CP=0.0052). This yielded a parsimonious tree balancing accuracy with interpretability.

2.4 K-Means Clustering

Unsupervised K-means clustering segmented patients into risk profiles without using disease labels. The Elbow Method indicated k=4 as optimal, and we characterized each cluster by demographic means and disease prevalence to assess clinical validity.

3. Results

3.1 Identifying Key Risk Factors

Both EDA and logistic regression converged on the same finding: **functional cardiac assessments outperform static measurements** as predictors. Three factor groups dominated:

1. **Chest Pain Type:** Asymptomatic (ASY) chest pain showed overwhelming disease prevalence, making it the single most powerful classifier.
2. **Exercise Stress Markers:** ST_SlopeFlat and ExerciseAngina are both indicators of ischemia under stress and were highly significant ($p<0.001$).
3. **Demographics:** Male sex remained a significant risk factor after controlling for clinical variables.

Notably, traditional risk factors (Cholesterol, RestingBP, Age) were *not statistically significant* when controlling for clinical markers, suggesting their predictive information is fully captured by functional assessments.

3.2 Model Performance

Model	AUC	Accuracy	Sensitivity	Specificity
Benchmark Logistic Reg.	0.940	88.7%	90.1%	87.0%
PCA-Enhanced Logistic Reg.	**0.945**	**89.1%**	**90.8%**	**87.0%**
Pruned Decision Tree	0.902	83.6%	84.2%	82.9%

Table 1. *Test set performance. The PCA-Enhanced model (highlighted) achieved the best discrimination while eliminating multicollinearity.*

The PCA-Enhanced model achieved the highest AUC (0.945) with balanced sensitivity and specificity, making it ideal for screening. The Decision Tree, while slightly less accurate, identified ST_SlopeUp as the optimal root split, patients with upward-sloping ST segments are immediately classified as low-risk, providing a simple clinical heuristic.

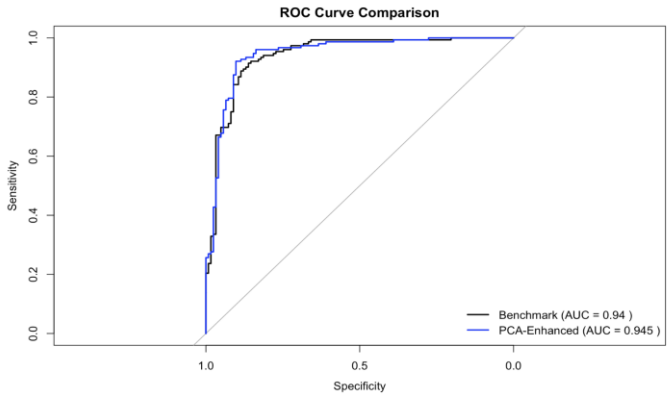
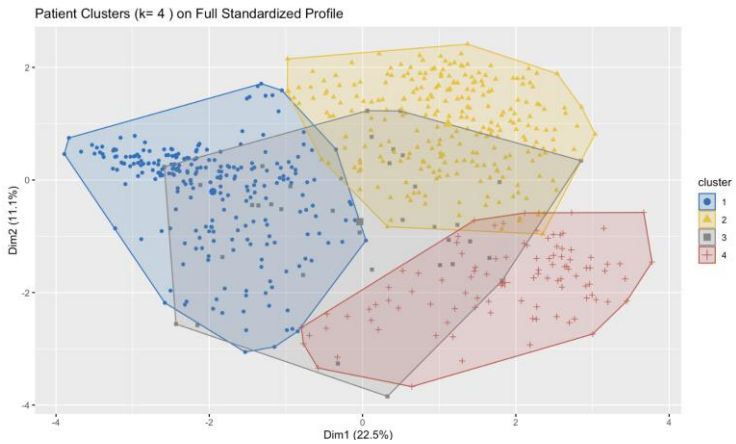


Figure 2. ROC Curve Comparison

3.3 Patient Segmentation via Clustering



K-means clustering validated our feature engineering by spontaneously separating patients into groups with dramatically different disease rates and without access to disease labels:

Figure 3. K-mean clusters of patients

- **Cluster 1 - 'Healthy Baseline' (15.5% prevalence):** Youngest patients (mean age 49), representing the low-risk reference group.
- **Cluster 2 - 'High-Risk' (84.0% prevalence):** Middle-aged patients with moderate cholesterol; risk driven by other factors.
- **Cluster 4 - 'Stealth Risk' (83.2% prevalence):** Oldest patients with paradoxically *lowest* cholesterol, likely diagnosed patients on statin therapy. This counterintuitive finding demonstrates that low cholesterol can signal existing disease management rather than low risk.

4. Discussion and Conclusions

4.1 Key Insights

This analysis yields three key insights for clinical practice. First, exercise stress markers are superior predictors with ST_Slope and ExerciseAngina provided stronger diagnostic signals than traditional risk factors, suggesting that functional cardiac assessments should be prioritized in screening protocols. Second, PCA improves model stability without sacrificing accuracy by transforming correlated features into orthogonal components, we achieved reliable coefficient estimates and marginally improved AUC. Third, unsupervised clustering can reveal clinical subgroups, the 'Stealth Risk' cluster demonstrates that identical biomarker values can have opposite clinical implications depending on patient context.

4.2 Model Recommendations

For maximum predictive accuracy, we recommend the PCA-Enhanced Logistic Regression (AUC=0.945). Its balanced sensitivity (90.8%) and specificity (87.0%) make it ideal for clinical screening where both false positives and false negatives carry significant costs. For clinical communication and transparency, the pruned Decision Tree provides interpretable rules that can be easily explained to patients and integrated into workflows, its root split on ST_SlopeUp offers a simple, actionable heuristic.

4.3 Limitations and Future Work

Our analysis is constrained to a single dataset without external validation. The sample may not generalize to all populations, and we did not account for potential confounders such as medication history or lifestyle factors. Future research should validate these models on independent cohorts, explore ensemble methods combining tree interpretability with regression accuracy, and investigate the clinical utility of the identified patient clusters for targeted intervention strategies.

5. References

fedesoriano. (2021). *Heart Failure Prediction Dataset* [Data set]. Kaggle. https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction